

Twin Pregnancy 05. Surveillance and Prenatal Diagnosis

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode five. Surveillance and prenatal diagnosis.

Source base

This episode is based on the Israeli twin follow-up position paper, Gabbe chapters on multiple gestation, genetic screening, and antepartum fetal evaluation, the NIPS position paper, the ultrasound position paper, and the local CVS and amniocentesis protocol.

Opening frame

Let us begin with the central idea.

Surveillance in twins is not more frequent singleton care. It is a schedule built around the complications predicted by chorionicity.

In dichorionic diamniotic twins, surveillance asks: are both fetuses growing, is there meaningful discordance, is either fetus small, is the cervix shortening, and is maternal disease emerging?

In monochorionic diamniotic twins, surveillance asks all of that, plus: is there early evidence of TTTS, TAPS, or sFGR?

In monochorionic monoamniotic twins, surveillance asks all of that, plus: how do we manage a cord-space risk that we cannot perfectly predict?

Start with DCDA follow-up

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The Israeli twin follow-up paper recommends first-trimester ultrasound, nuchal translucency, second-trimester anomaly scan, and growth scans every 4 weeks from 24 weeks in uncomplicated DCDA twins.

The logic is that DCDA twins have independent placentas or at least independent chorions. They can still have growth problems, preterm birth, preeclampsia, and fetal demise, but they do not require every-two-week TTTS surveillance.

Growth discordance matters. It is commonly calculated as the estimated weight difference divided by the larger twin weight. A difference above about 20 to 25 percent is clinically important in the Israeli paper. The risk increases further when one fetus also meets criteria for growth restriction.

Now MCDA follow-up

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The Israeli paper recommends ultrasound follow-up starting from 16 to 18 weeks for MCDA twins. The exam should assess amniotic fluid in both sacs, image both fetal bladders, follow growth every 2 to 4 weeks, and include umbilical artery and middle cerebral artery Dopplers. Fetal echocardiography is recommended around 20 to 24 weeks.

Why these elements?

Fluid and bladder assessment screen for TTTS. Growth and umbilical artery Dopplers screen for sFGR and placental territory problems. Middle cerebral artery Doppler helps screen for TAPS. Fetal echocardiography matters because recipient cardiac dysfunction may appear in TTTS, and monochorionic pregnancy itself is a cardiac-risk context.

Now MCMA follow-up

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The Israeli paper emphasizes increased risk from cord complications and notes debate about whether hospitalization and monitoring from 28 weeks reduce mortality. Delivery is recommended between 32 and 34 weeks. The local protocol begins considering inpatient monitoring and daily monitoring earlier, from 24 to 26 weeks, and recommends cesarean delivery at 32 to 33 weeks after steroids, with magnesium if delivery is before 32 weeks.

This is a good example of guideline logic. When a risk is severe but hard to predict, management often shifts from detection to planned preterm delivery at a gestational age where neonatal risk is acceptable.

Now prenatal genetic screening

Now prenatal genetic screening.

Twin pregnancy complicates screening because the pregnancy contains either two genetically distinct fetuses or two genetically identical fetuses, depending on zygosity. In dizygotic twins, one fetus can be normal and the other aneuploid. Serum marker

signals can be diluted or blunted. Gabbe notes that serum screening is less sensitive in twins than in singleton pregnancy.

Nuchal translucency is therefore especially important. It gives fetus-specific information, and in Israel it is the screening test of choice for Down syndrome in multiple pregnancies. The second trimester can include maternal serum alpha fetoprotein, or AFP, or ultrasound to screen for open neural tube defects.

Now NIPS, or non-invasive prenatal screening

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The NIPS position paper states that NIPS can be performed from 10 weeks in singleton and twin pregnancies, and that performance for common trisomies in twins is broadly similar to singleton pregnancy. But the limitations are essential.

In twin pregnancies, NIPS does not assess sex chromosome abnormalities or submicroscopic abnormalities in the same way. If a Y chromosome is detected, one or both fetuses is likely male. It does not tell you which fetus. If no Y chromosome is detected, both are likely female.

A pregnancy that started as twins and underwent spontaneous or induced reduction is not recommended for NIPS because residual cell-free DNA from the demised or reduced fetus can cause false positive results. This is not a lab nuisance. It is a direct consequence of placental DNA biology.

Triplet and higher-order data are insufficient.

Now invasive diagnosis

Now invasive diagnosis.

Amniocentesis and chorionic villus sampling, or CVS, require careful fetal labeling. Before sampling, document chorionicity, placental location, fetal viability, anatomy, fetal sex if known, sac position, and any distinguishing structural marker.

For amniocentesis in multiple pregnancy, all fetuses can usually be sampled. Techniques may include separate punctures or a single puncture crossing the membrane, depending on operator choice and anatomy. The local protocol warns about contamination, dead space, sampling the same sac twice, and the need to visualize membrane tenting and release if crossing the septum.

CVS is different. In dichorionic twins, both placentas generally need sampling. In monochorionic twins, because they are usually genetically identical, one sample may be enough if there are no significant sonographic differences. The local protocol states that CVS in multiple pregnancy is performed with two punctures rather than a single puncture approach.

The practical reason to prefer early diagnostic clarity is that selective reduction, if needed, is safer earlier than later. But that statement must always be balanced with the couple's values, the fetal findings, chorionicity, and procedure risk.

Now antepartum fetal testing

Now antepartum fetal testing.

Twin pregnancies have higher perinatal mortality than singletons, and risk rises with chorionicity complexity. But testing is not a substitute for correct timing of delivery. Nonstress tests, biophysical profiles, Dopplers, growth scans, and fluid assessment each answer different questions. A reassuring test today does not erase the accumulated risk that determines delivery timing.

Pause and summarize.

DCDA surveillance focuses on growth, discordance, cervix, maternal disease, and usual fetal assessment. MCDA surveillance adds screening for TTTS, TAPS, and sFGR beginning around 16 to 18 weeks. MCMA surveillance adds cord-risk management and planned early cesarean delivery. Genetic screening and invasive testing require fetus-specific labeling and careful counseling.

Now final synthesis

Now final synthesis.

Key point

First, surveillance follows chorionicity.

Second, in twins, screening tests often lose fetus-specific clarity. That is why nuchal translucency, detailed ultrasound, and careful labeling matter.

Third, NIPS is useful in twins for common trisomies, but reduced or vanishing twin pregnancies are a major false-positive trap.

Fourth, invasive testing in twins is a procedural identification problem as much as a needle problem. Know exactly which fetus and which sac or placenta you are sampling.

End of episode five.